CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 7: Ratio and Proportion

Every formula and result of the chapter on one sheet

1	Duplicate ratio of $a : b$	$a^2 : b^2$
2	Triplicate ratio of $a : b$	$a^3 : b^3$
3	Sub-duplicate ratio of $a : b$	$\sqrt{a} : \sqrt{b}$
4	Sub-triplicate ratio of $a : b$	$\sqrt[3]{a} : \sqrt[3]{b}$
5	Reciprocal ratio of $a : b$	$b : a$
6	Compound ratio of $a : b$ and $c : d$	$ac : bd$
7	Proportion $a : b = c : d$	$ad = bc$
8	Continued proportion $a : b = b : c$	$b^2 = ac$
9	Mean proportional between a and c	$\sqrt{ac}$
10	Third proportional to a and b	$\frac{b^2}{a}$
11	Fourth proportional to a, b, c	$\frac{bc}{a}$
12	Componendo	$\frac{a+b}{b} = \frac{c+d}{d}$
13	Dividendo	$\frac{a-b}{b} = \frac{c-d}{d}$
14	Componendo and dividendo	$\frac{a+b}{a-b} = \frac{c+d}{c-d}$